ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** increase in girth	Secondary growth.
2	**(b)** squamous epithelium	Alveoli.
3	**(b)** hinge joint	One plane.
4	**(b)** vascular cambium	Rings.
5	**(d)** sclerenchyma	Dead and strong.
6	**(b)** trachea	Moves mucus.
7	**(a)** cartilage and synovial fluid	Lubrication.
8	**(a)** secondary xylem	Seasonal growth.
9	**(a)** collenchyma	Corner thickening.
10	**(c)** bone	Rigid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They are actively dividing (1); Dense cytoplasm and thin walls (1).	1+1
12	Phloem is removed (1); Food cannot reach the roots (1).	1+1
13	Bone is hard; cartilage is flexible (1); Bone has Ca; cartilage has no blood vessels (1).	1+1
14	Xylem vessels: dead, hollow, lignified (1); Sieve tubes: living, with sieve plates, no nucleus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water enters, guard cells swell (1); Stoma opens (1); Water leaves, they become flaccid and the stoma closes (1).	1+1+1
16	Plants: xylem and phloem (1); Animals: blood and lymph (1); Plants have meristems; animals have dividing cells everywhere (1).	1+1+1
17	Tissues: epithelial, muscle, connective (1); Organ: stomach (1); System: digestive (1).	1+1+1
	OR: Simple: one type of cells, e.g. parenchyma (1); Complex: more than one type, e.g. xylem (1); Complex conduct materials (1).	1+1+1
18	Reduces water loss (1); Epidermis secretes it (1); Adaptation (1).	1+1+1
	OR: Branched cells with intercalated discs (1); Rhythmic contraction without fatigue (1); Abundant mitochondria (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Thin section, stain and mount (1); Epidermis: protection (1); Cortex: storage and support (1); Vascular bundles: conduction (1); Pith: storage (1).	1+1+1+1+1
	OR: Aquatic: aerenchyma (1); Desert: thick cuticle, sunken stomata (1); Land: strong xylem (1); Root tissues (1); Examples (1).	1+1+1+1+1
20	Xylem and phloem together in a bundle (1); Stem: on the same radius (1); Root: on alternate radii (1); Xylem towards the centre (1); Phloem towards the outside (1).	1+1+1+1+1
	OR: Apical meristem: length (1); Lateral meristem: girth (1); Intercalary: grasses (1); Differentiation (1); Annual rings (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.

- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** cartilage	Flexible pads.
2	**(c)** cork and phloem	Outer layers.
3	**(c)** phloem	Sieve tubes.
4	**(a)** plasma	Fluid.
5	**(b)** insulation	Adipose.
6	**(c)** supportive tissues in plants and muscular tissue in animals	Different tissues.
7	**(c)** xylem vessels	Dead at maturity.
8	**(c)** haemoglobin	Pigment.
9	**(a)** elastic and connect bones	Elastic.
10	**(d)** food downward	Phloem is in the bark.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Not under our will (1); Cross striations; branched cells (1).	1+1
12	Smooth: spindle-shaped, unstriated, involuntary (1); Skeletal: striated, voluntary (1).	1+1
13	Cambium cells divide (1); Add secondary xylem and phloem (1).	1+1
14	(a) Squamous (1); (b) Columnar; (c) ciliated columnar (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Made of different cell types (1); They form vascular bundles (1); Conduct water and food (1).	1+1+1
16	Epidermis: protection (1); Mesophyll: photosynthesis (1); Vascular bundles: conduction (1).	1+1+1
17	Dendrite to cell body (1); Cell body to axon (1); Axon terminals to the next neuron or muscle (1).	1+1+1
	OR: Appearance (1); Control (1); Location (1).	1+1+1
18	Cells divide at different rates (1); Epithelium regenerates quickly (1); Nerve tissues do not (1).	1+1+1
	OR: Sclerenchyma (1); Collenchyma (1); Turgid parenchyma (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Squamous (1); Columnar (1); Blood (1); Bone (1); Cartilage (1).	1+1+1+1+1
	OR: Hinge (1); Ball and socket (1); Pivot (1); Fixed (1); Role of cartilage and ligaments (1).	1+1+1+1+1
20	Voluntary: limbs (1); Involuntary: gut, heart (1); Voluntary under the will (1); Involuntary by autonomic system (1); Structure differences (1).	1+1+1+1+1
	OR: Plasma: transport (1); RBC: haemoglobin (1); WBC: defence (1); Platelets: clotting (1); Blood as a connective tissue (1).	1+1+1+1+1

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- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** xylem	Evaporation from leaves.
2	**(d)** fighting germs	Defence.
3	**(a)** carries impulses away from the cell body	Output.
4	**(c)** protective	Protect the tip.
5	**(d)** aerenchyma	Buoyancy.
6	**(c)** clotting of blood	Stops bleeding.
7	**(a)** receive impulses	Input.
8	**(a)** parenchyma	Storage.
9	**(a)** chloroplasts	Only guard cells have them.
10	**(c)** cardiac muscle	Works continuously.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They make sugar by photosynthesis (1); Sugar helps them take in water and open the stoma (1).	1+1
12	Support sieve tubes (1); Control their activity (1).	1+1
13	More space for haemoglobin (1); More oxygen can be carried (1).	1+1
14	Provides buoyancy (1); Stores gases for exchange (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Skin: protection (1); Intestine: absorption (1); Lungs: exchange of gases (1).	1+1+1
16	Long thin extension (1); Large surface area for absorption (1); Thin wall (1).	1+1+1
17	Tissue: similar cells (1); Organ: several tissues (1); Muscle and heart (1).	1+1+1
	OR: Bone: matrix with Ca and P (1); Cartilage: flexible matrix (1); Bone supports; cartilage cushions (1).	1+1+1
18	Suberin deposited in walls (1); Prevents water loss (1); Protects the stem (1).	1+1+1
	OR: Tissues are transparent (1); Stains highlight parts (1); Safranin stains lignin red (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Blood clots at the wound (1); Epithelial cells divide to cover (1); Bone cells (osteoblasts) form new bone (1); Connective tissue forms scar (1); Time needed (1).	1+1+1+1+1
	OR: Epidermis transparent (1); Palisade packed with chloroplasts (1); Spongy has air spaces (1); Veins bring water, take food (1); Stomata for gases (1).	1+1+1+1+1
20	Receives stimuli (1); Conducts impulses (1); Sensory, relay, motor neurons (1); Reflex arc (1); Rapid response (1).	1+1+1+1+1
	OR: Micropropagation of plants (1); Disease-free stock (1); Skin grafts (1); Stem cell therapy (1); Ethical aspects (1).	1+1+1+1+1

Mistakes to avoid:

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